

Baby Sofia · large-for-gestational-age birth complicated by shoulder dystocia · right arm adducted, internally rotated, elbow extended, forearm pronated ("**waiter's tip**") · MRI: **traumatic injury to C5-C6 roots of the brachial plexus** · **Erb-Duchenne palsy**

BIO 004 · WEEK 8 · CASE DEEP DIVE

The Nerve Plexuses

A newborn whose right arm hangs limp at her side in the classic waiter's tip posture. A traumatic delivery. A pair of nerve roots torn from the spinal cord.

PATIENT CASE

Baby Sofia was a 4.4-kg infant born vaginally after a difficult delivery complicated by **shoulder dystocia** — her right shoulder caught against her mother's pubic symphysis. The obstetric team used downward traction on her head to deliver the shoulder.

In the newborn nursery she does not move her right arm. The arm hangs at her side: **adducted, internally rotated, elbow extended, forearm pronated, wrist flexed** — the classic "**waiter's tip**" posture. Hand grip on the right is preserved. MRI confirms injury to the **upper trunk of the brachial plexus (C5-C6 roots)**. The diagnosis is **Erb-Duchenne palsy**. Physical therapy is initiated; ~80% of mild cases recover spontaneously within a year.

A plexus is what happens when the simple "one spinal nerve, one body region" arrangement isn't enough. The PNS reshuffles the nerve fibers of multiple spinal nerves into mixed bundles that then become the named peripheral nerves of the limbs.

GUIDING QUESTION

What is a plexus, why do the limbs need them, and how does an injury to two specific roots produce a stereotyped posture across the whole arm?

1 WHY PLEXUSES

Reshuffling for distal precision

Most ventral rami in the trunk (T2-T12 intercostals) keep their segmental identity — one nerve per intercostal space. But the limbs need precise, mixed innervation: muscles in the upper limb get fibers from C5-T1, not from any single segment. So in the limb regions, several ventral rami merge into a **plexus**, mix and reshuffle fibers, and emerge as named peripheral nerves carrying contributions from multiple segments.

Four major plexuses:

Cervical (C1-C4)

Neck muscles; sensation from neck and shoulder; the *phrenic nerve* (C3-C5) to the diaphragm.

Brachial (C5-T1)

Entire upper limb. *Sofia's*.

Lumbar (L1-L4)

Anterior thigh and medial leg; the *femoral nerve*, *obturator nerve*.

Sacral (L4-S4)

Posterior thigh, leg, foot, perineum; the *sciatic nerve* (and its tibial/common peroneal branches), *pudendal nerve*.

Roots, Trunks, Divisions, Cords, Branches

The brachial plexus is the most clinically tested plexus. Mnemonic for its organization: "*Real Texans Drink Cold Beer*" — Roots → Trunks → Divisions → Cords → Branches.

Roots (5)

Ventral rami of C5, C6, C7, C8, T1.

Trunks (3)

Upper (C5+C6 — *Sofia's injury*), *Middle* (C7),
Lower (C8+T1).

Divisions (6)

Each trunk splits into anterior and posterior divisions.

Cords (3)

Lateral (anterior divisions of upper + middle trunks), *Medial* (anterior division of lower trunk),
Posterior (all three posterior divisions).

Branches (5 major)

Musculocutaneous (lateral cord), *Median* (lateral + medial cords), *Ulnar* (medial cord), *Axillary* (posterior cord), *Radial* (posterior cord).

What each major nerve does

NERVE	MOTOR	SENSORY
Musculocutaneous	Biceps, brachialis, coracobrachialis (elbow flexion, forearm supination)	Lateral forearm
Median	Most forearm flexors; thenar muscles	Palmar thumb-index-middle, half of ring finger
Ulnar	Most intrinsic hand muscles; some forearm flexors	Little finger, half of ring finger
Axillary	Deltoid (shoulder abduction); teres minor	Lateral shoulder
Radial	All forearm extensors; triceps (elbow extension); wrist/finger extension	Dorsum of hand (lateral)

Lower limb innervation



Femoral nerve (L2-L4)

Anterior thigh muscles (quadriceps — knee extension); skin of anterior thigh and medial leg (saphenous).



Obturator nerve (L2-L4)

Medial thigh adductors; skin of medial thigh.



Sciatic nerve (L4-S3)

Largest nerve in body. Hamstrings + all muscles below knee. Divides into *tibial* (posterior leg + sole of foot) and *common peroneal* (anterolateral leg + dorsum of foot).



Pudendal nerve (S2-S4)

Perineum and external genitalia. Target of pudendal block in obstetrics.

Beyond Erb

INJURY	ROOTS	FINDINGS
Erb-Duchenne (upper trunk)	C5-C6	Waiter's tip. <i>Sofia</i> .
Klumpke (lower trunk)	C8-T1	Claw hand; sometimes Horner's syndrome (T1 sympathetic).
Sciatica	L4-S3 compression	Pain radiating down posterior leg; often from disc herniation.
Cauda equina syndrome	multiple lumbar/sacral roots	Saddle anesthesia, bowel/bladder dysfunction. Surgical emergency.

6 WHY WAITER'S TIP

C5-C6 motor loss adds up to one posture

The classic Erb posture comes from loss of specific muscles:

- **Deltoid + supraspinatus paralyzed** (axillary, C5) → no shoulder abduction → arm *adducted*.
- **Infraspinatus + teres minor paralyzed** → unopposed internal rotators → arm *internally rotated*.
- **Biceps + brachialis paralyzed** (musculocutaneous, C5-C6) → no elbow flexion → arm *extended at elbow*.
- **Supinator paralyzed** → unopposed pronators → forearm *pronated*.
- **Wrist flexors intact** (C7-T1) → wrist falls into *flexion*.

The hand itself is largely spared (C7-T1 root function intact). Sofia can grip — only the proximal arm posture is abnormal.



DRAW ZONE 1 -- Sketch the brachial plexus. Label roots, trunks, divisions, cords, and the five major branches.

Mark C5-C6 (Sofia's injury) and trace where it goes — musculocutaneous + axillary + part of median/radial.



DRAW ZONE 2 -- Draw Sofia's waiter's-tip posture. Label which muscle losses cause each component: shoulder adduction, internal rotation, elbow extension, forearm pronation, wrist flexion.

Spared hand grip distinguishes Erb (upper trunk) from Klumpke (lower trunk).

Two roots, one stereotyped posture

Erb-Duchenne palsy is plexus anatomy boiled down to two roots. Every component of Sofia's arm posture maps to a specific muscle group losing innervation.

"Shoulder dystocia → upper-trunk traction injury"

Forceful downward traction on the head while the shoulder is impacted stretches the angle between neck and shoulder. The upper trunk (C5-C6) takes most of the stretch; it can be neuropraxic (transient), axonotmetic (axon disruption), or neurotmetic (complete tear).

"Arm adducted, elbow extended"

Deltoid (abduction) and biceps/brachialis (flexion) are out. The arm hangs limp.

"Internally rotated, forearm pronated"

External rotators (infraspinatus, teres minor) and supinator are out. Unopposed internal rotators and pronators take over.

"Hand grip preserved"

C7-T1 (median and ulnar roots) are intact. Distinguishes Erb from Klumpke palsy (C8-T1 injury), which produces claw hand with preserved shoulder.

"Prognosis"

~80-90% of newborn Erb palsies recover within 12 months with PT to prevent contractures. Severe cases (root avulsion vs stretch) may need nerve grafting or tendon transfers.

"Why phrenic was safe"

Phrenic nerve (C3-C5) comes off the CERVICAL plexus, not the brachial. Sofia's diaphragm function is intact.

Root, trunk, cord, branch. Sofia's posture is the brachial plexus telling you exactly which two roots took the force.

